

Ethanol Blending in Gasoline – Nigeria

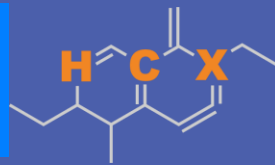
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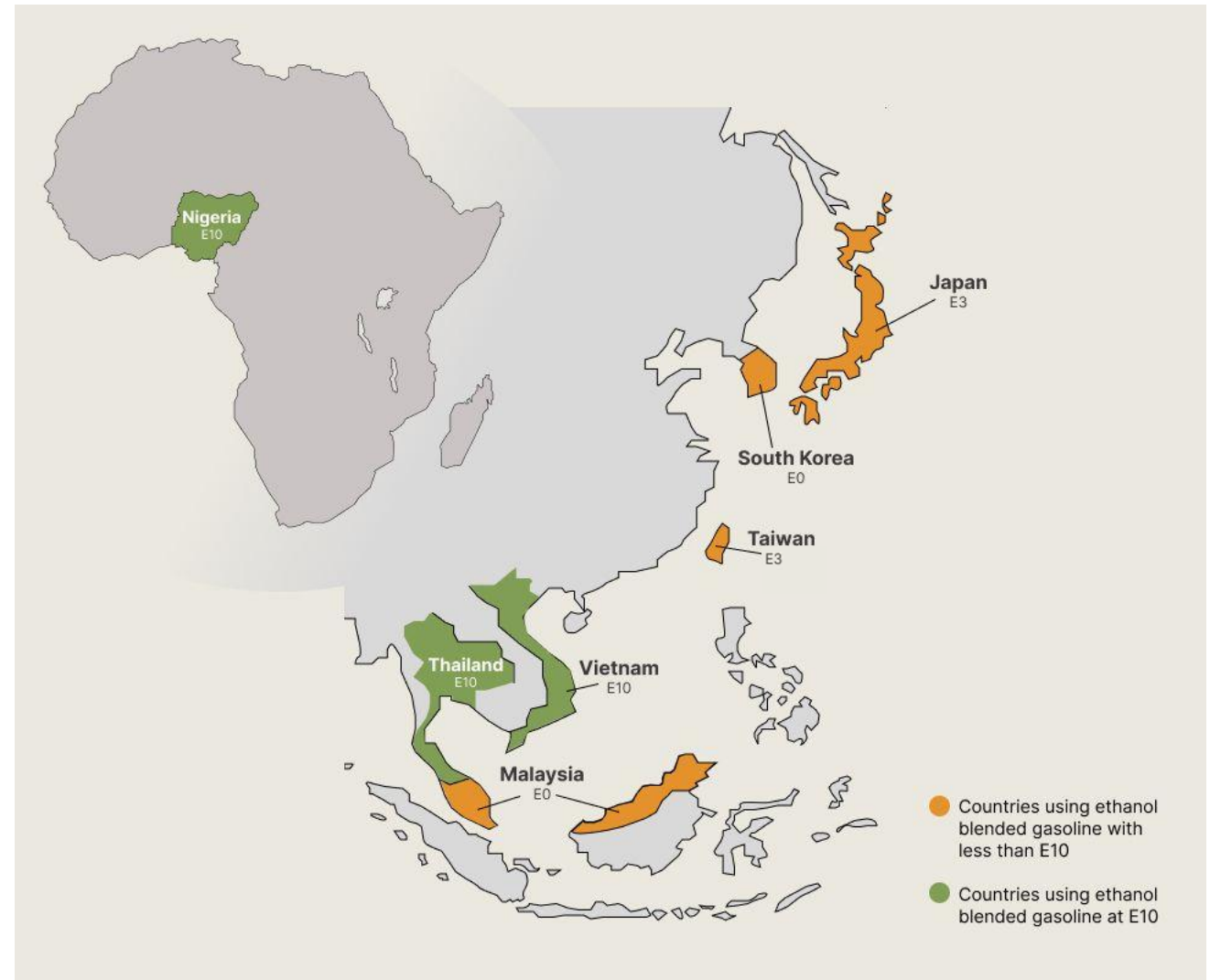


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Ethanol Blending in Asia-Africa

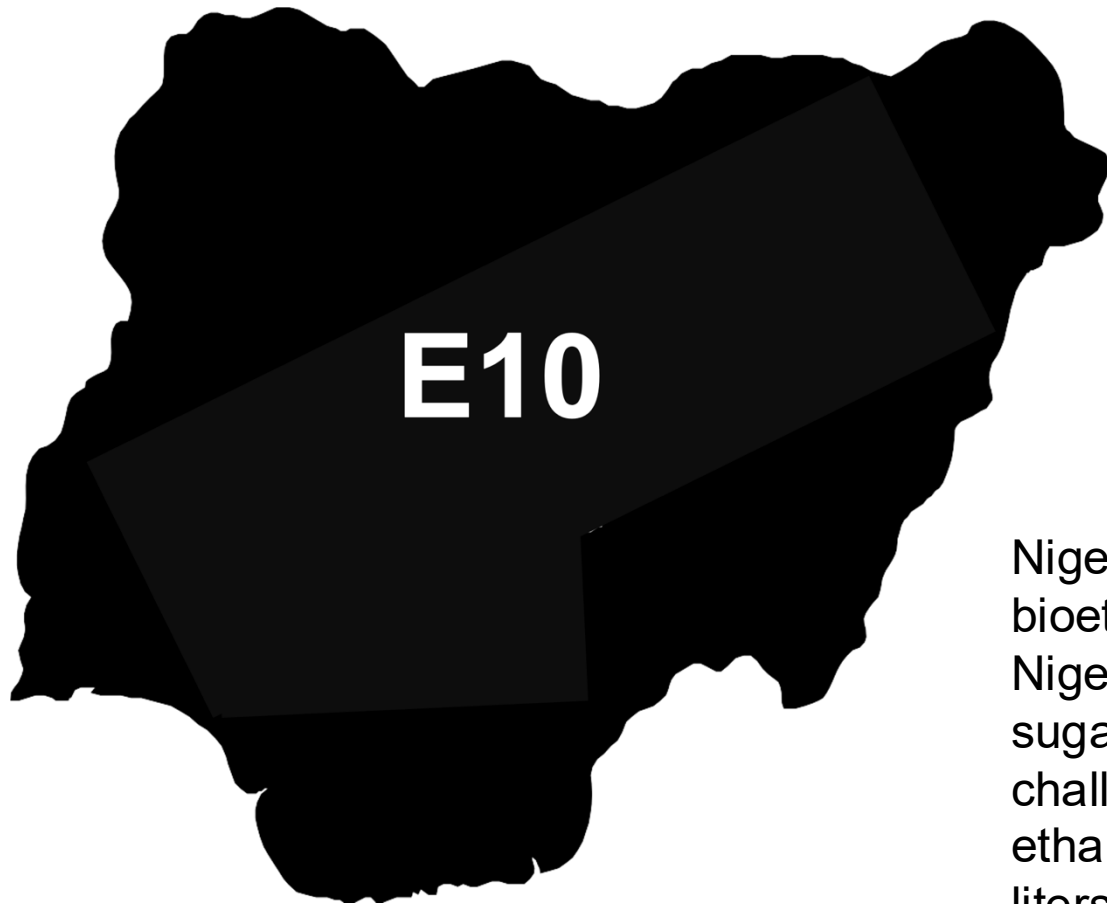
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Seven countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

Ethanol Gasoline Blending in Nigeria

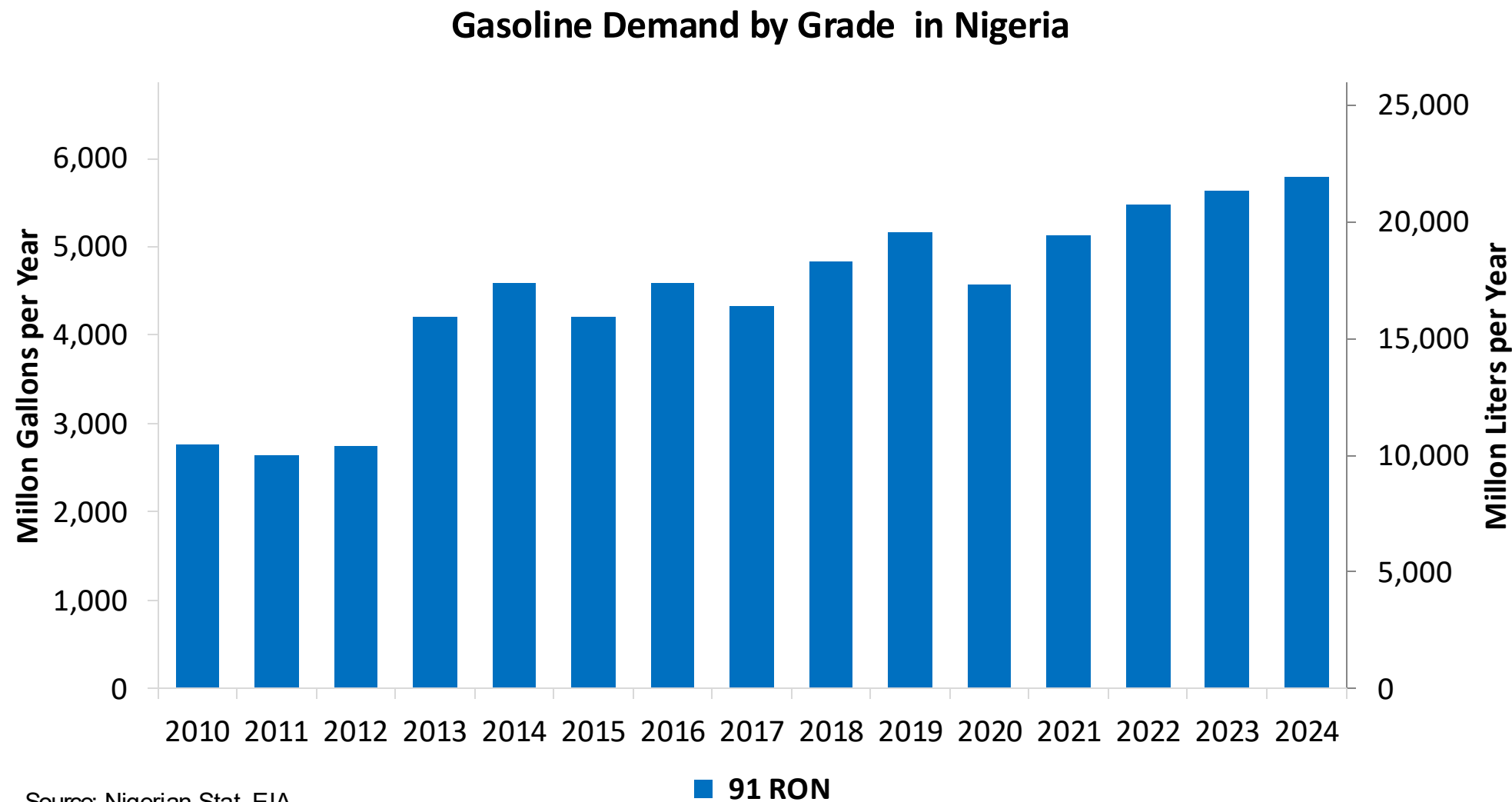


In 2024, gasoline consumption in Nigeria was 21,894 million liters (5,784 million gallons) and growth rate was 2.3%. Gasoline production and net imports were 9,242 and 12,652 million liters (2,441 and 3,342 million gallons) correspondingly. Nigeria offers one main grade: RON 91. Production and supply of a 95 RON grade will be commercially available in the short term.

Nigeria has a Biofuels Policy that aims for a 10% blend of bioethanol in gasoline, but this target has not been met. Nigeria produces bioethanol primarily from cassava and sugarcane, but the industry is small-scale and faces challenges, resulting in the country being a net importer of ethanol. Ethanol consumption in Nigeria was 379 million liters (100 million gallons) representing 1.7% of gasoline demand.

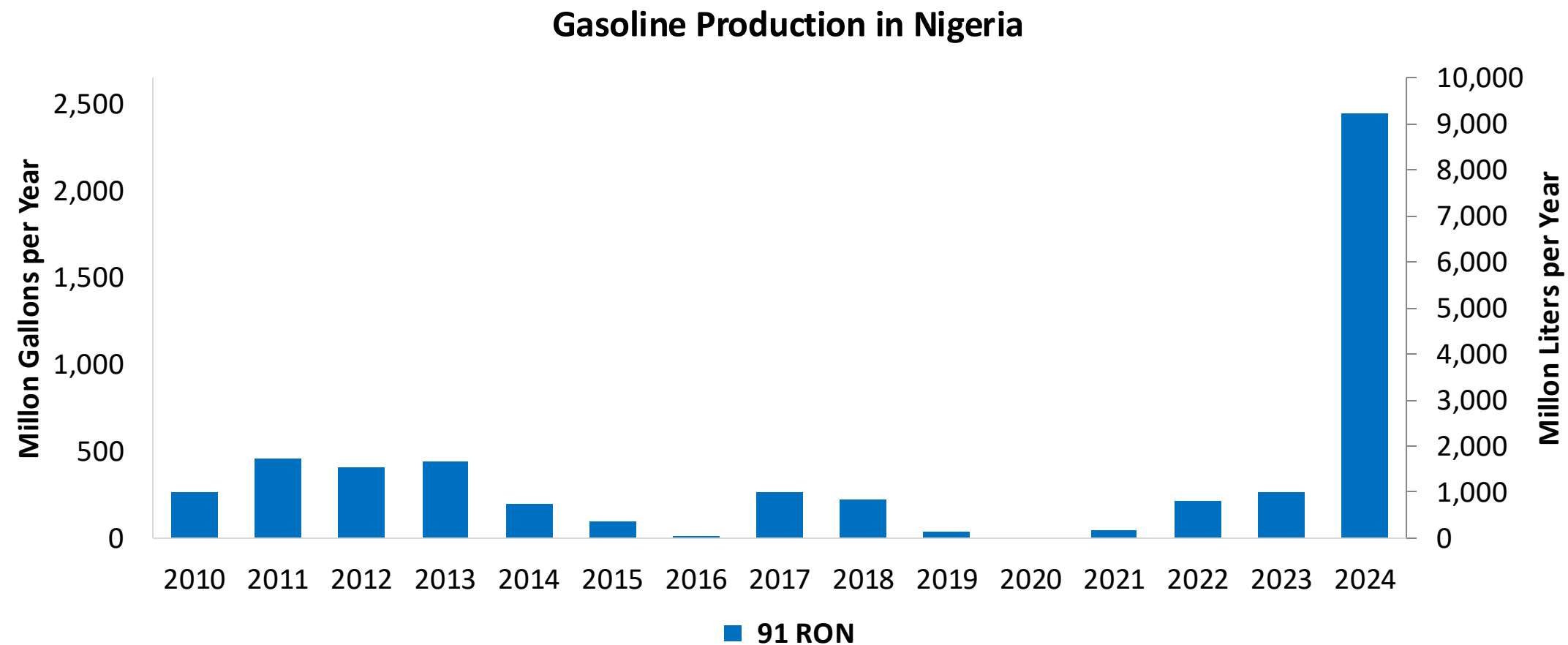
Source: Nigerian Stat, EIA, Climate Laws, Advanced Biofuels

Gasoline Demand in Nigeria



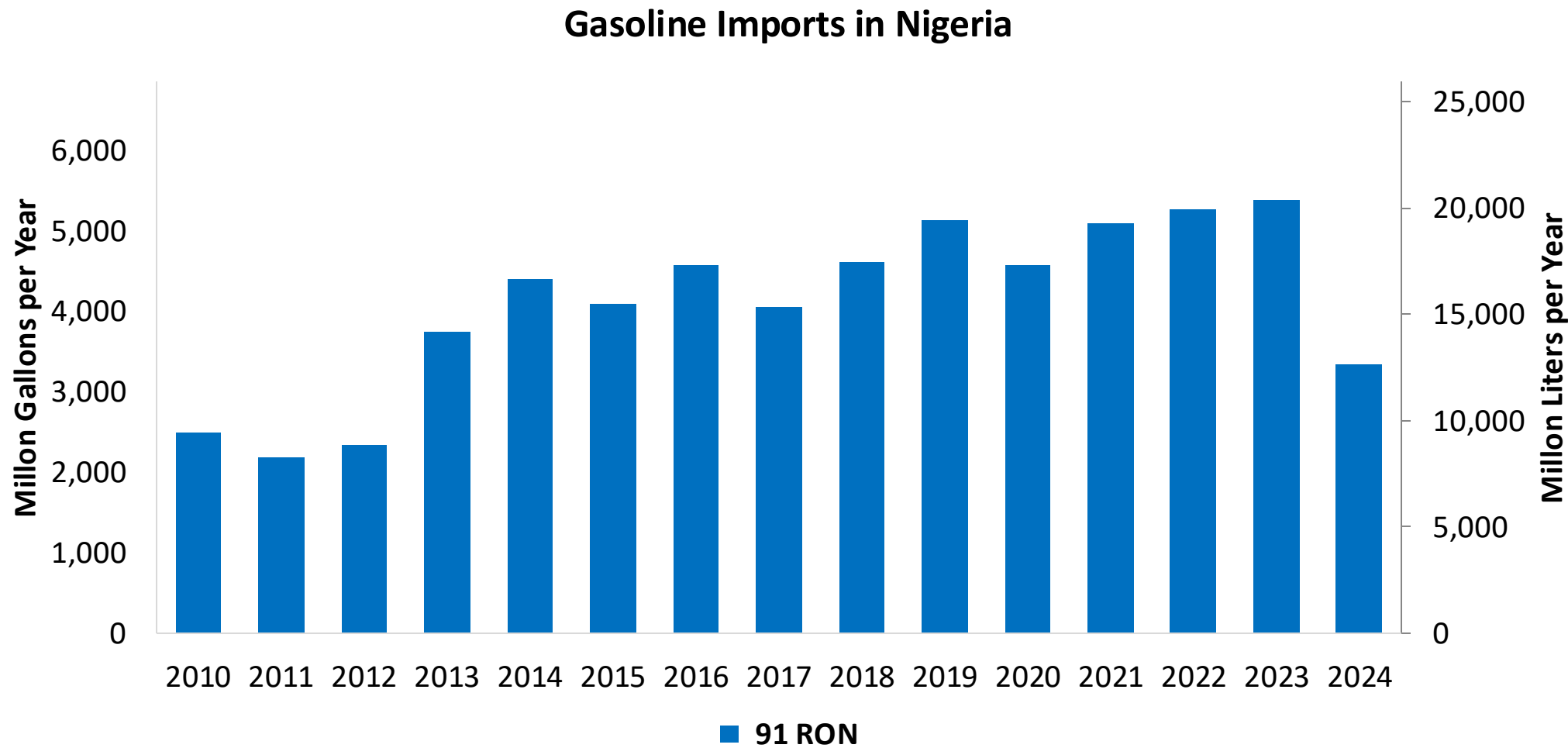
Source: Nigerian Stat, EIA

Gasoline Production in Nigeria



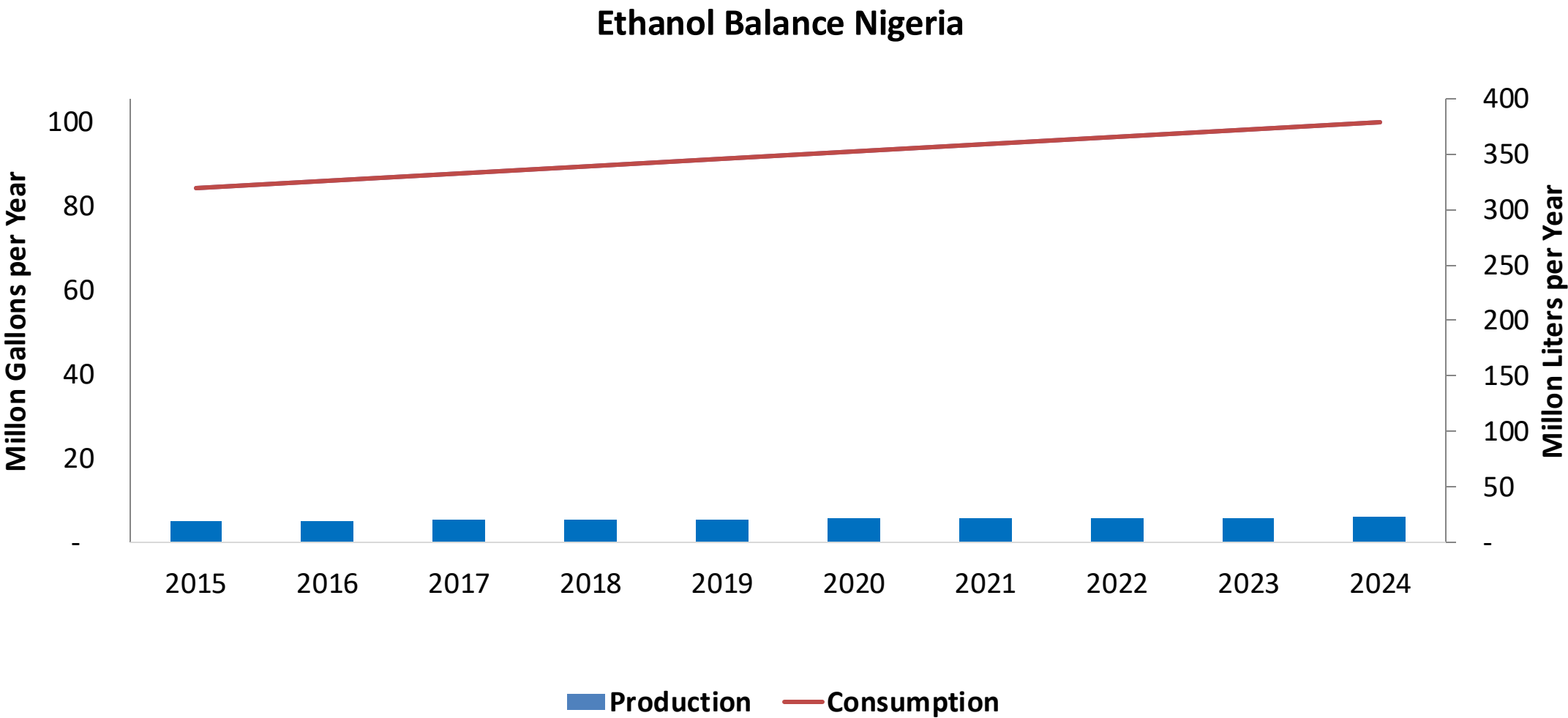
Source: Nigerian Stat, EIA

Gasoline Imports in Nigeria



Source: Nigerian Stat, EIA

Ethanol Balance in Nigeria



Source: Nigerian Stat, Climate Laws, Advanced Biofuels

Gasoline Quality in Nigeria

Name	NIS116:2017. E10 (NIS 534:2006)			
Implementation Year	2017			
Applicability	Nationwide			
Grade	PMS		PMS E10	
	Min	Max	Min	Max
RON	91.0		92.0	
MON	81.0		82.0	
AKI				
Benzene (%vol)		2.0		2.0
Aromatics (% vol)				
Olefins (%vol)				
Lead (g/L)		0.002		0.002
Manganese (mg/L)		-		-
Sulfur (ppm)		150		150
Oxygen (%v)		0.2		3.5
Ethanol (%vol)				10.0
MTBE (%vol)				
RVP (kPa)		62.0		69.0

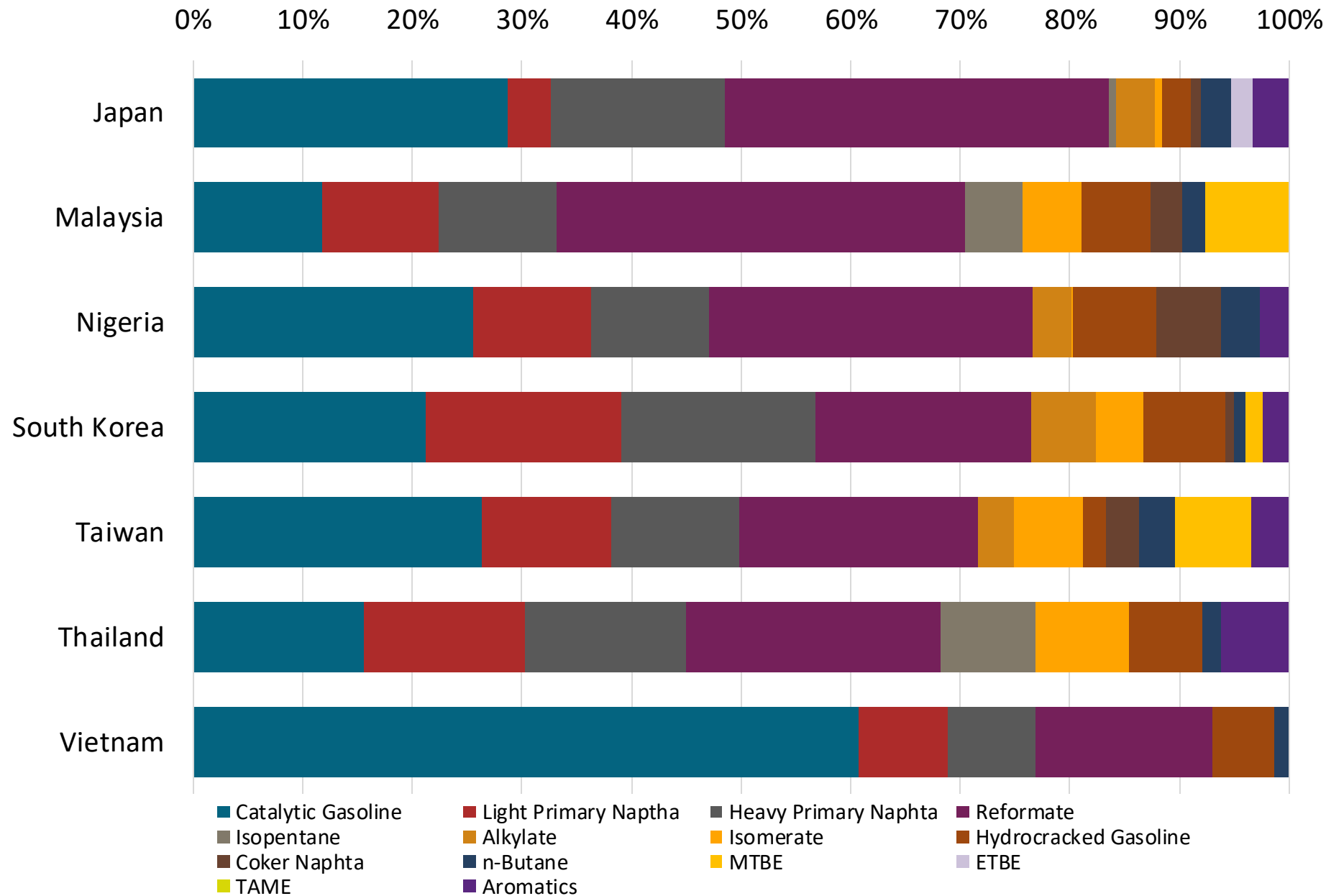
Source: US Grains Council

Gasoline Component Blending in Asia Africa

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Asia Africa are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source: Faro90



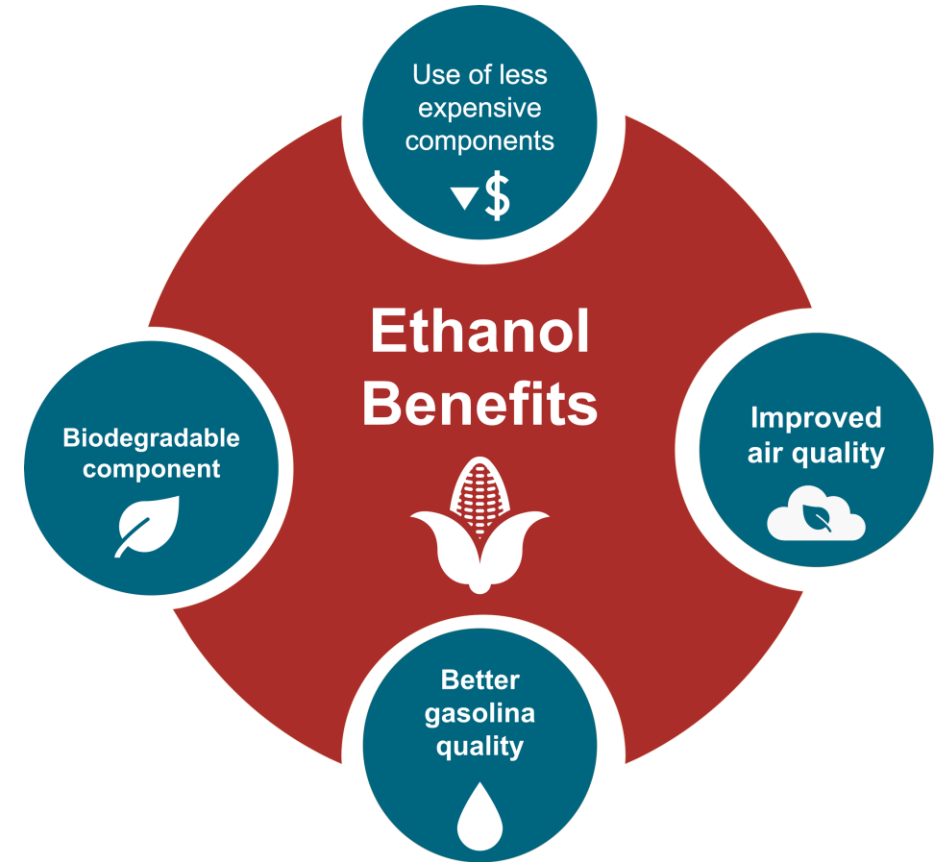
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

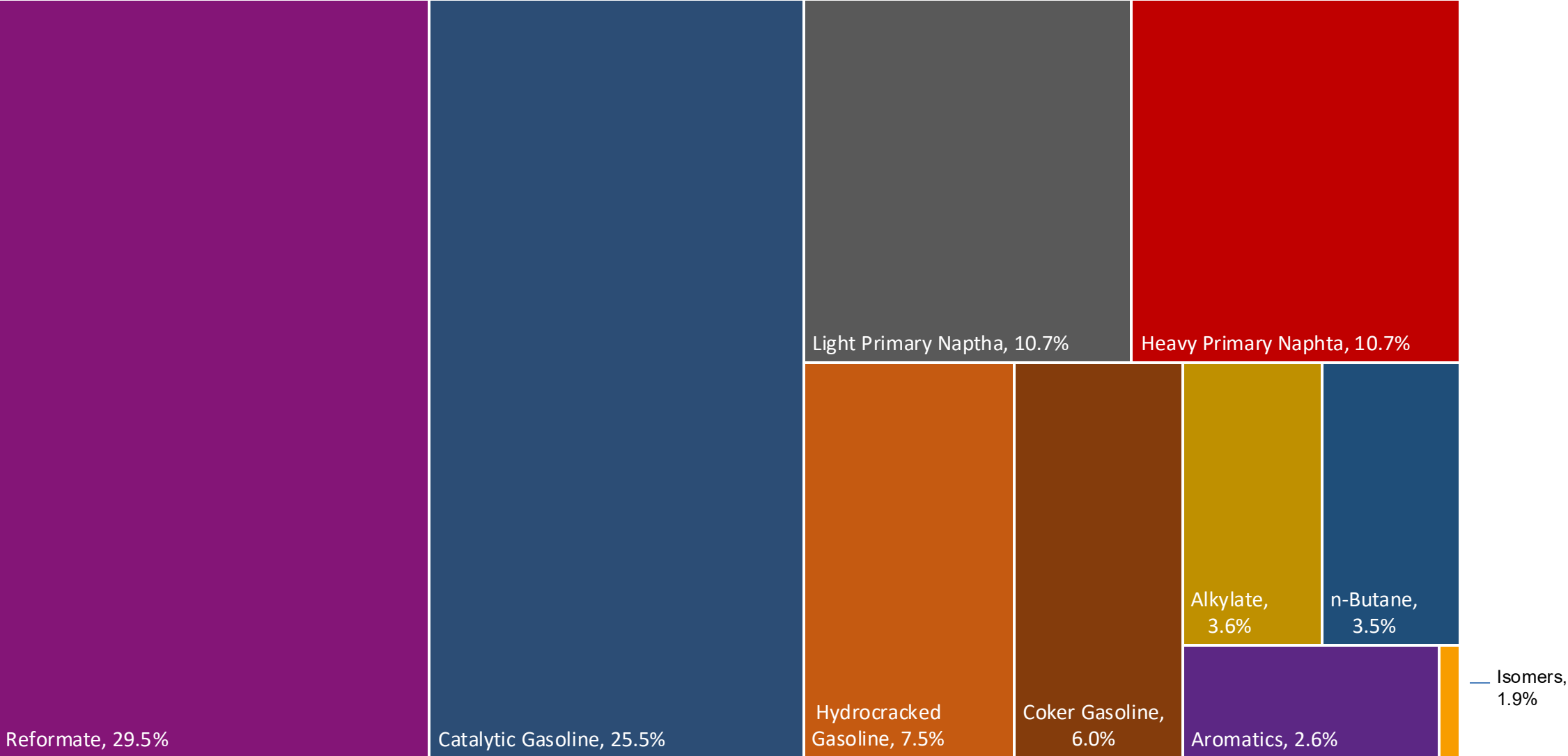
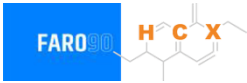
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

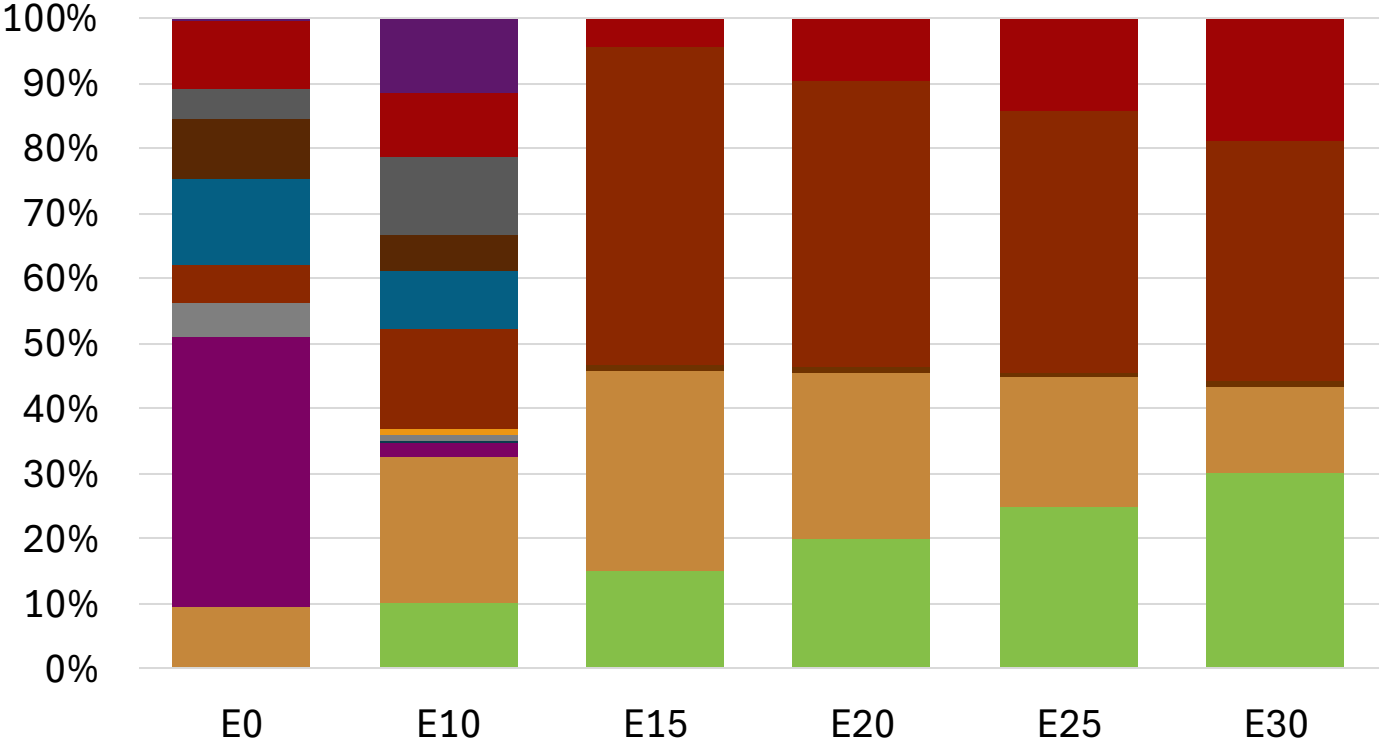
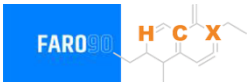


Nigeria Available Blending Components



Source: Faro90

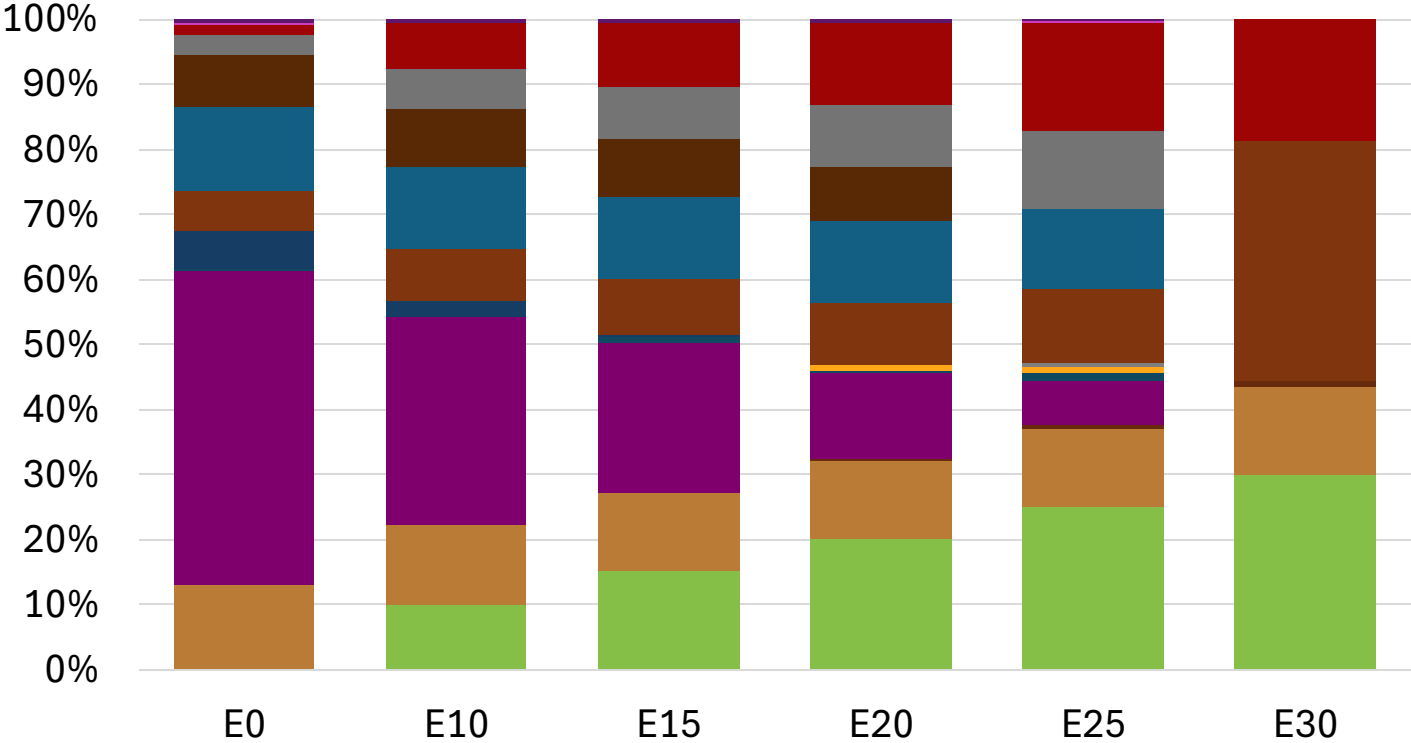
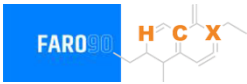
Nigeria – Gasoline 91 – Constant Octane



Average CIF 2024
Prices

Source: Faro90

Nigeria - Gasoline 95 - Constant Octane

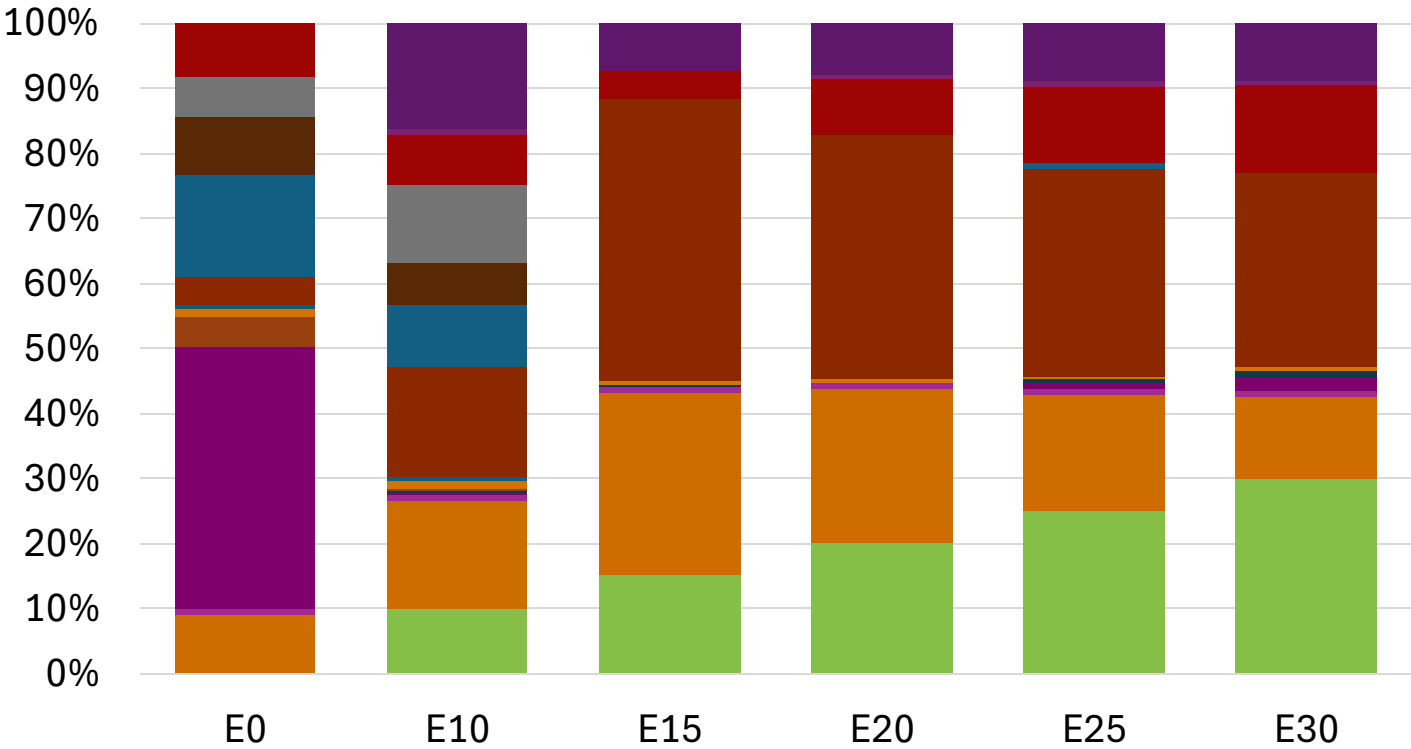


Average CIF 2024 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.2
Price (US\$/gal)	\$ 2.416	\$ 2.293	\$ 2.220	\$ 2.142	\$ 2.057	\$ 2.005

Source: Faro90

Nigeria – Gasoline 91 – Increased Octane

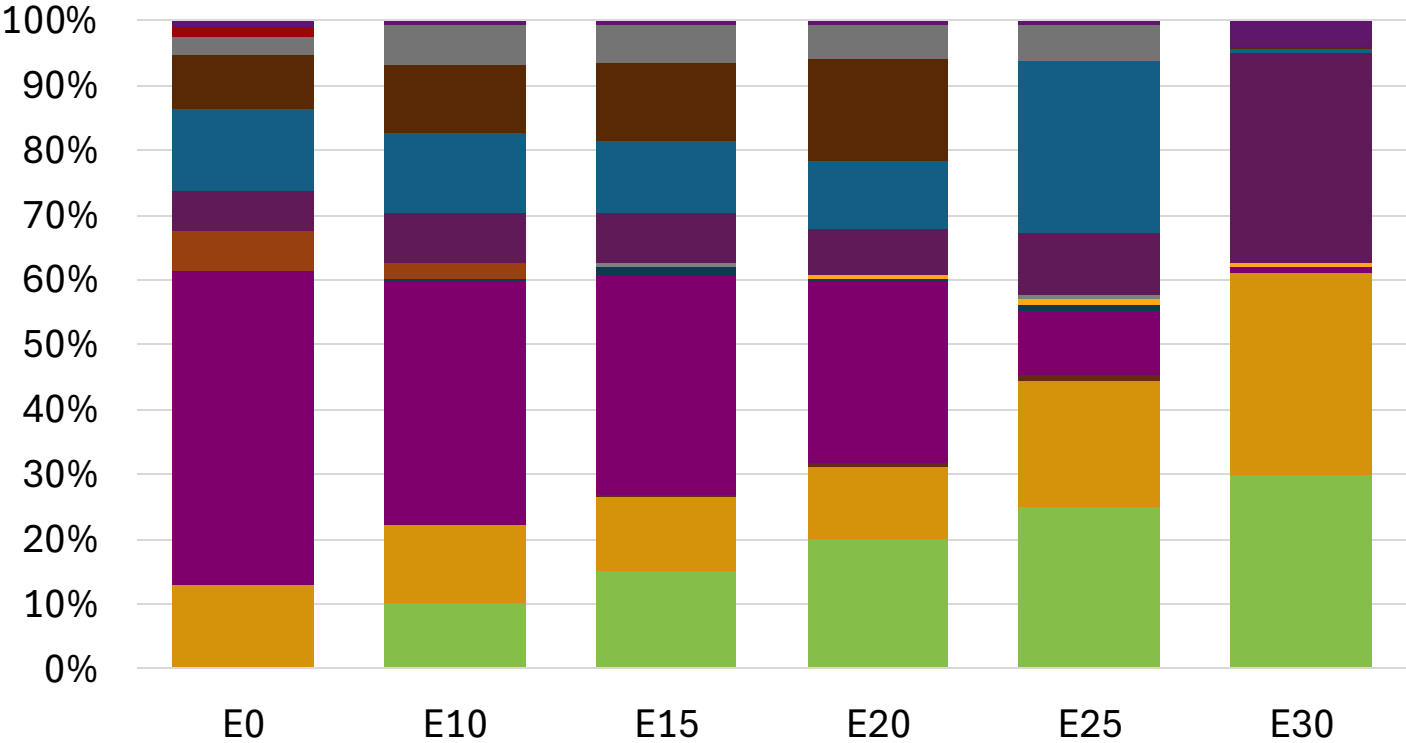


Average CIF 2024
Prices

Octane (RON)	91.0	91.6	92.8	94.6	96.8	98.6
Price (US\$/gal)	\$ 2.230	\$ 2.230	\$ 2.230	\$ 2.230	\$ 2.230	\$ 2.230

Source: Faro90

Nigeria – Gasoline 95 – Increased Octane



Average CIF 2024
Prices

Octane (RON)	95.0	97.6	99.0	100.4	100.9	101.0
Price (US\$/gal)	\$ 2.416	\$ 2.416	\$ 2.416	\$ 2.416	\$ 2.416	\$ 2.416

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

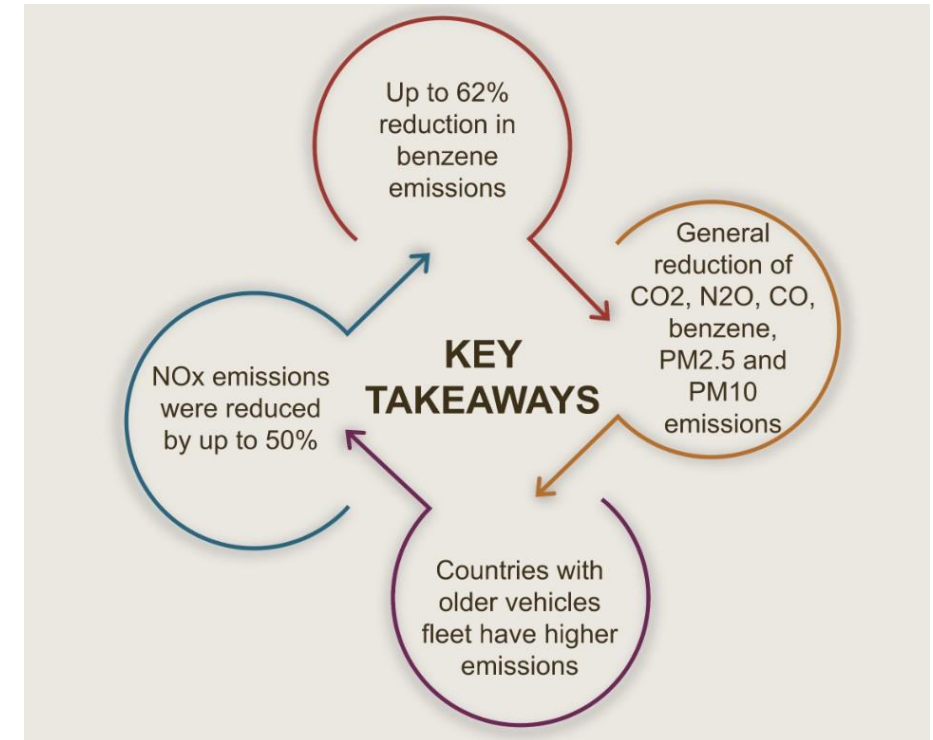
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

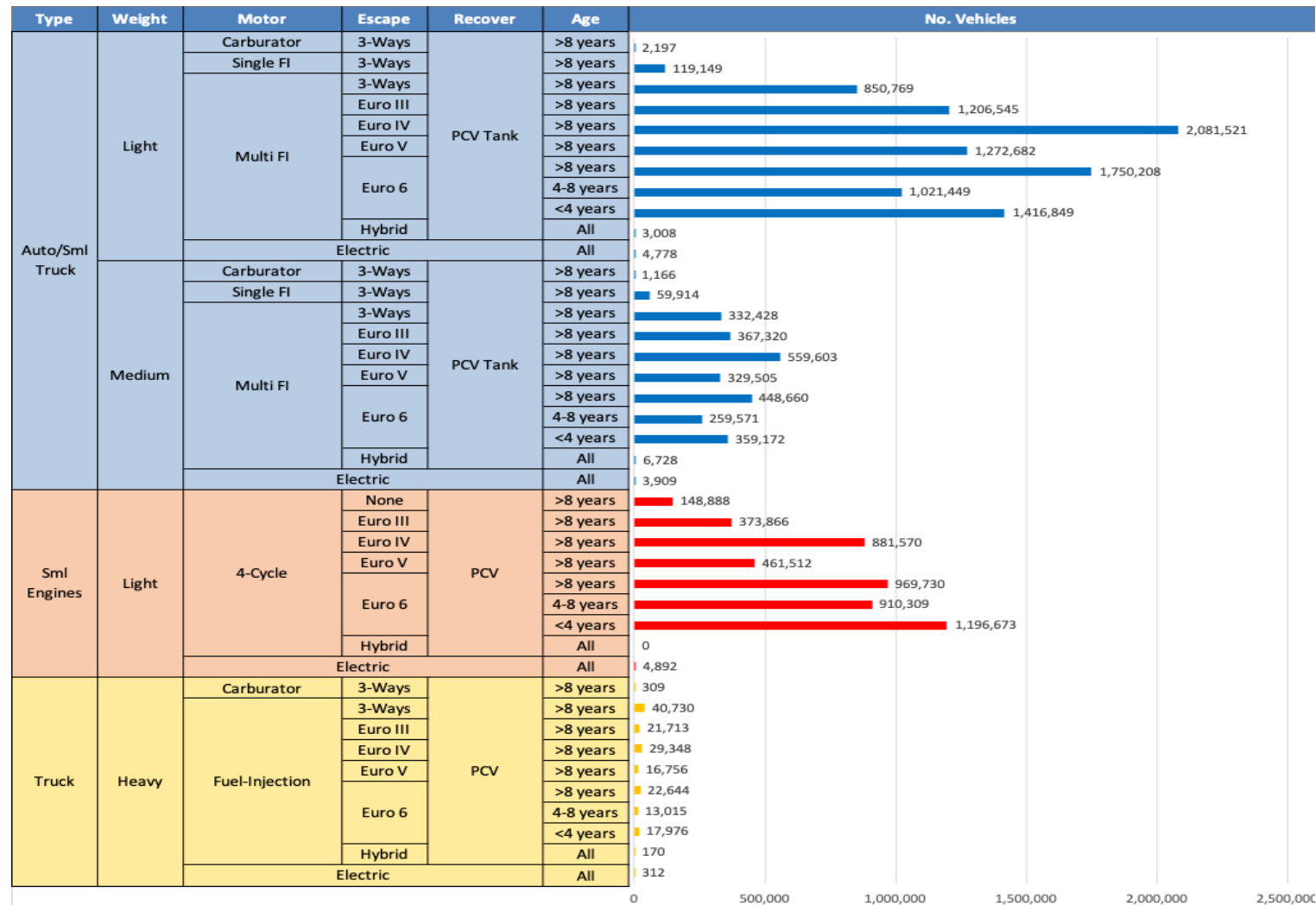
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Nigeria

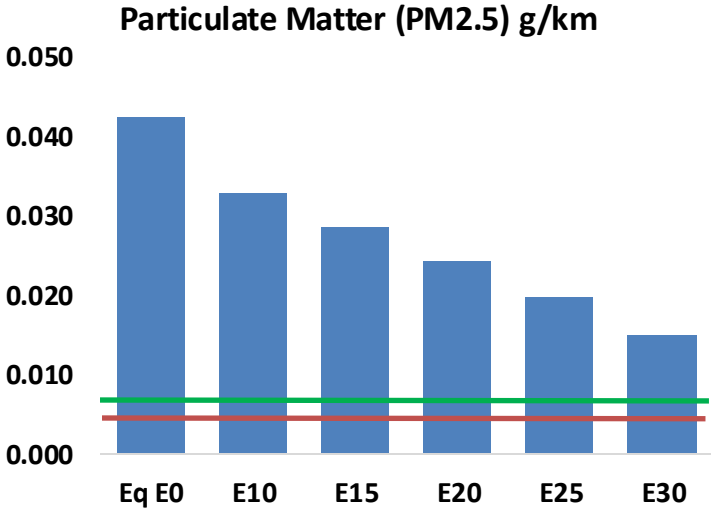
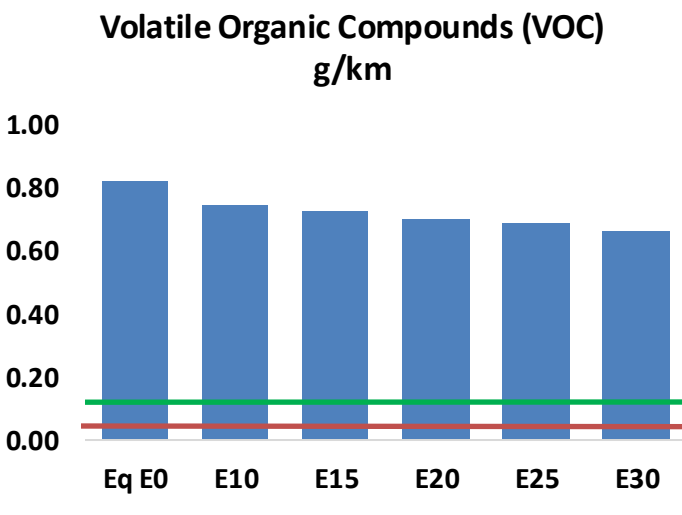
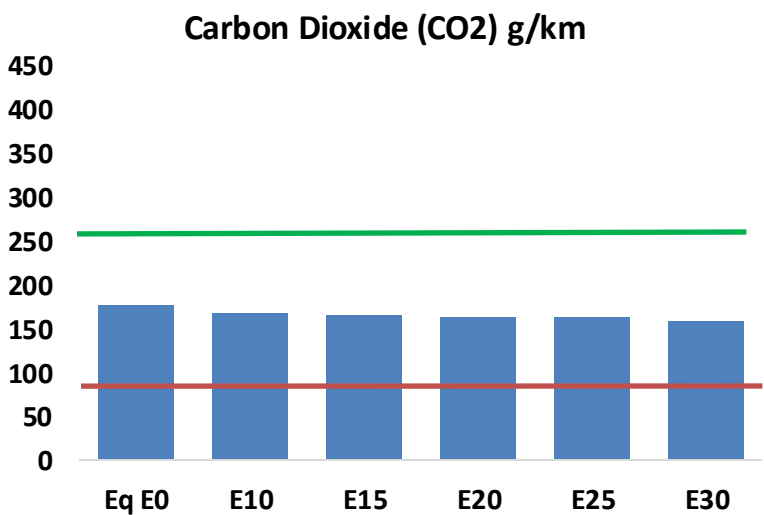
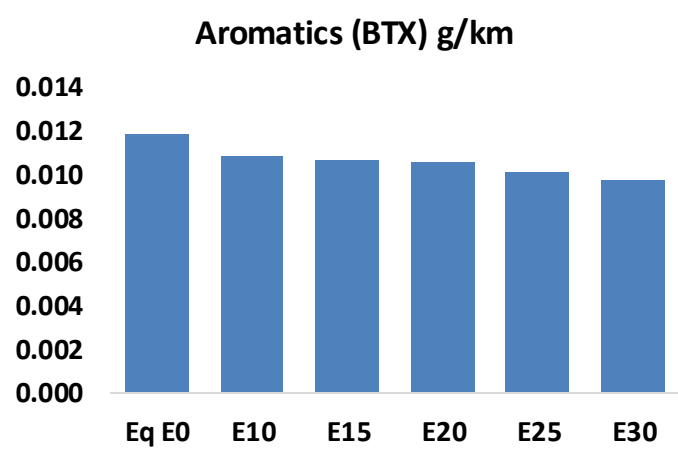
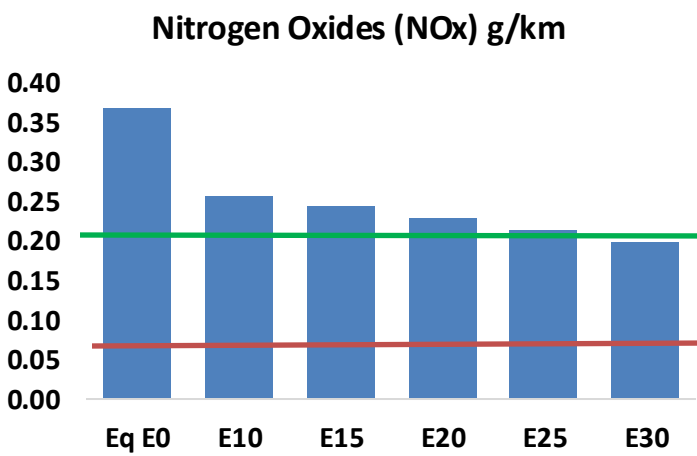
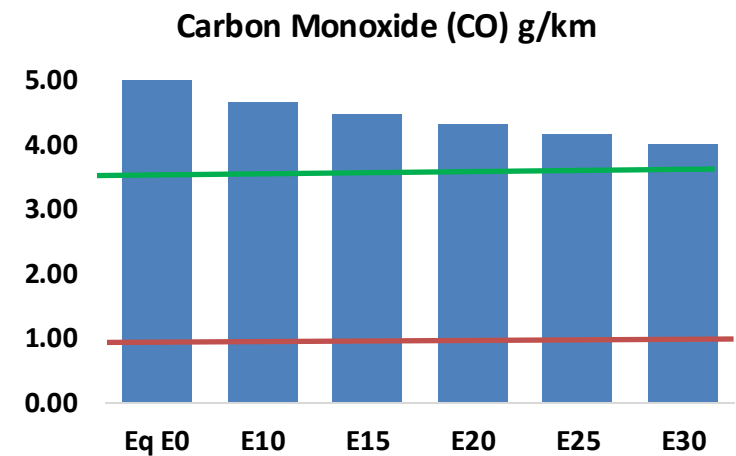
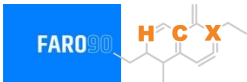


Vehicle Fleet: 17,567,546
Gasoline Vehicle Fleet: 16,203,007

Source: Nigeria Stat

Nigeria – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



Nigeria – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,051,399	1,948,219	1,845,039	1,771,126	1,701,643	1,649,693	1,581,391	-10%	-17%
CO2	70,013,767	68,213,413	66,513,078	65,175,815	64,514,085	64,188,202	63,005,028	-5%	-8%
VOC	323,008	308,955	294,902	285,489	276,985	270,738	261,371	-9%	-14%
NOx	145,686	109,264	101,980	96,116	90,646	84,614	78,233	-30%	-38%
SOx	3,421	3,163	2,904	2,686	2,476	2,248	2,009	-15%	-28%
NH3	28,328	28,047	27,767	27,899	28,212	28,433	28,617	-2%	0%
N2O	1,245	1,238	1,233	1,239	1,265	1,279	1,293	-1%	2%
CH4	70,376	70,376	70,376	71,432	72,980	74,317	75,584	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,053	1,985	1,917	1,879	1,849	1,828	1,792	-7%	-10%
Acetaldehyde	5,652	6,487	9,518	14,494	19,747	22,564	26,662	68%	249%
Formaldehyde	21,184	20,595	24,008	28,191	29,534	32,672	35,568	13%	39%
Benzene	4,692	4,509	4,272	4,205	4,170	3,988	3,859	-9%	-11%
PM2.5	4,605	4,089	3,573	3,103	2,635	2,138	1,620	-22%	-43%
PM10	12,129	10,770	9,411	8,172	6,940	5,631	4,267	-22%	-43%

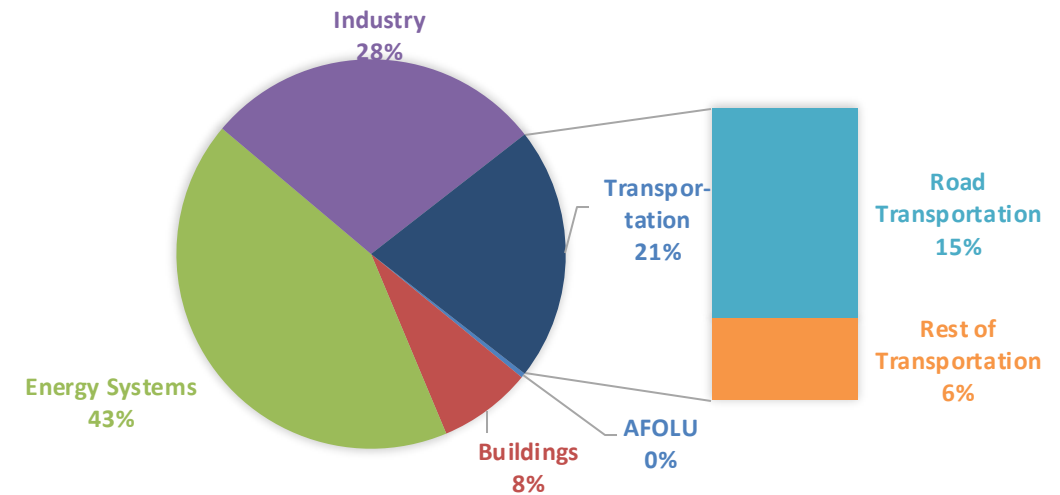
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

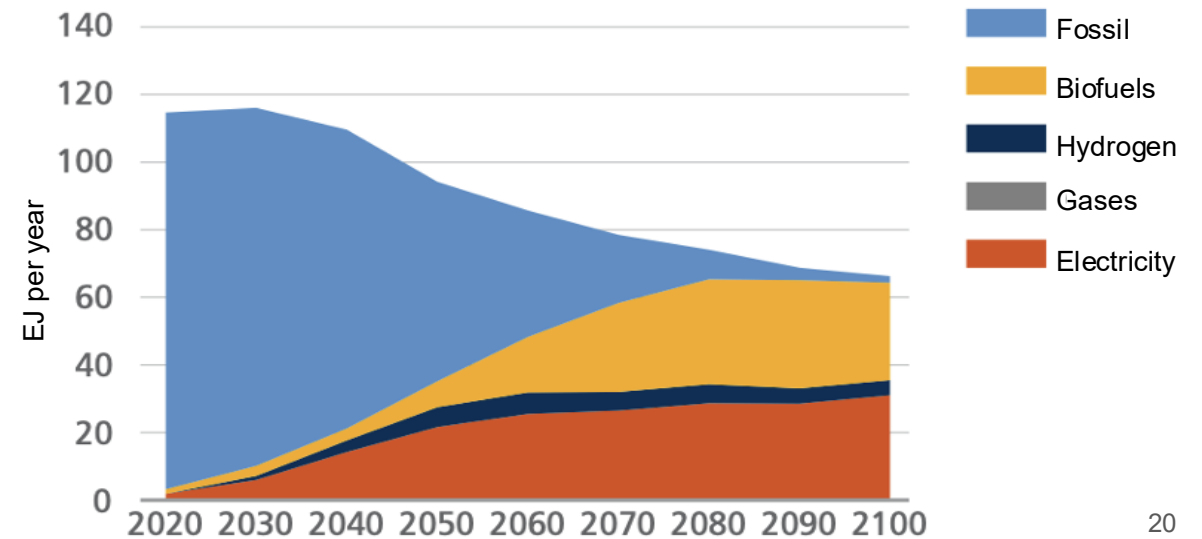
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

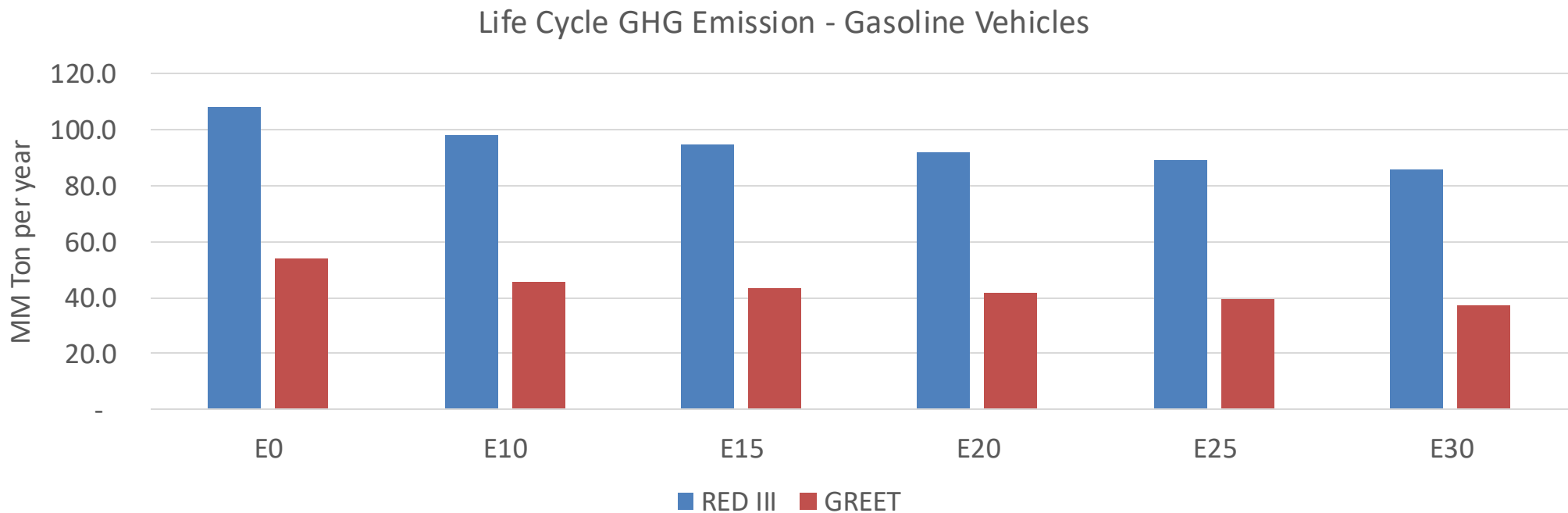
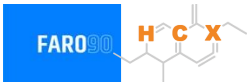
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Nigeria – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	413.6
Target @ 2035 (MMT/year)	267.2
Delta	146.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	15.0%	19.1%	22.8%	26.4%	30.7%
RED II/III	0.0%	9.3%	12.3%	14.9%	17.5%	20.4%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.6%	7.1%	8.4%	9.7%	11.3%
RED II/III	0.0%	6.9%	9.0%	11.0%	12.9%	15.0%

Source: Greet, RED II/III, climateactiontracker.org, F90